

DATA PRODUCT CARD

PJATK • Big Data Course • ecommerce_marketing_pipeline

1. PRODUCT OVERVIEW

Product name: ecommerce_marketing_pipeline
Owner / Team: Alicja Tomaszewska, s35133
Business domain: Ecommerce/Marketing
Primary users: Marketing teams, E-commerce managers, Data analysts, Researchers, Students
Version / update date: v1.0 / 2026-05-03

AT A GLANCE

Format: Table (DuckDB), plus CSV input, optional Kafka stream
Refresh: Manual / event-driven (sensor + schedule in Dagster)
Access: GitHub repository (public)
Quality metrics: Completeness, Uniqueness (event_id), Freshness
Status: Production-like

2. PURPOSE & VALUE

What problem does this data product solve? What decision, analysis, or use case does it support?

This data product solves the problem of inconsistent, manually calculated e-commerce metrics (views, purchases, revenue, conversion).

Event data is automatically ingested into the bronze layer (from CSV files or Kafka), cleaned and modeled in silver, and aggregated into the gold layer (marketing_report).

The product supports marketing analysis and business decisions, for example **identifying which brands and categories generate the highest revenue and conversion over time**.

Thanks to built-in data quality metrics and dbt tests, users can quickly assess whether the data is trustworthy.

This allows to connect to a ready-to-use mart and build visualizations without needing additional explanations.

3. DATA SOURCES

Source	Type	Refresh
data/raw/*.csv (event exports)	File (CSV)	On demand / when file appears
Kafka topic ecommerce-events	Streaming queue	Configurable (e.g. every 5 min schedule)
data/fixtures/ecommerce_sample.csv (smoke/sample)	File (CSV)	Manual

4. HOW TO ACCESS THIS PRODUCT

- Where is the product published?** GitHub repository (public)
- Direct link:** https://github.com/AlicjaTomaszewska/dagster-dbt-duckdb-pipeline/tree/master/data/marketing_report/
- How to download or query it:** download *marketing_report.csv* file from the link above
- Credentials or access notes:** not required – public repository
- Example file/table name:** *marketing_report.csv*

5. QUALITY SNAPSHOT

Metric	Value
Completeness	silver_fact_events_completeness_key_cols_pct (threshold: >= 95%); obtained value: 100%
Uniqueness	silver_fact_events_event_id_uniqueness_pct (threshold: 100%); obtained value: 100%
Row count / freshness	silver_fact_events_row_count – obtained value: 42418543; silver_fact_events_freshness_hours – obtained value: 0.0167

These quality metrics indicate that the dataset is currently highly reliable: key-column completeness is 100%, event ID uniqueness is 100% (no detected duplicates), and the latest silver-layer update is very recent (freshness = 0.0167 hours, about 1 minute). The row count (42,418,543) confirms that the dataset is large and non-trivial, so analytical results should be considered production-like for course use. Freshness is measured as the time since the latest_loaded_at in fact_events; lower values mean more up-to-date data.

6. HIGH-LEVEL FLOW

Summarize the pipeline in one line:

Source	Bronze	Silver	Gold	Product
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CSV files / Kafka topic -> bronze_raw_events (bronze) -> dim_products + dim_users + fact_events (silver) -> marketing_report + dq_pipeline_metrics (gold) -> MS Teams Data Marketplace

Detailed Contract & Operating Information

7. SCHEMA OVERVIEW

Field name	Type	Description	Example
report_date	DATE	Calendar date (UTC-based) for the aggregated events in this row.	2019-10-02
brand	VARCHAR	Product brand for the aggregated group. Missing/blank values are normalized to __UNKNOWN__.	samsung
category_l1	VARCHAR	Top-level product category (first segment of category path). Missing values are normalized to __UNKNOWN__.	electronics
total_views	BIGINT	Number of view events for the given date, brand, and top category.	12540
total_purchases	BIGINT	Number of purchase events for the same aggregation grain.	842
revenue	DOUBLE	Total purchase value (sum of purchase prices) for the aggregation grain.	154230.50
conversion_rate_pct	DOUBLE	Conversion rate in percent: $100 * \text{total_purchases} / \text{total_views}$. Can be NULL when total_views = 0.	6.71

_refreshed_at	TIMESTAMP	Timestamp when this aggregated row was last recomputed by the pipeline. Useful for freshness checks.	2026-05-03 13:23:23.926052+02
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8. REFRESH & OPERATIONS

Refresh frequency: Manual / near real-time (schedule)

Update trigger: Dagster sensor (new_csv_file_sensor) + Dagster schedule (kafka_ingestion_schedule)

Storage location: DuckDB file - ecommerce_warehouse.duckdb, marketing_report (gold) exported to CSV file

Expected batch size / volume: [e.g. 5,000 rows/day]

Dependencies: file drop CSV / Kafka / dbt / DuckDB / Dagster

9. EXAMPLE USAGE

Business question:

Which brands and top-level categories generate the highest revenue over time, and how do that metric change day by day?

Example query / code / steps:

```
import pandas as pd

df = pd.read_csv("marketing_report.csv")

top_segments = (

    df.sort_values(["report_date", "revenue"],
ascending=[False, False])

    .head(10)

)

print(top_segments[[

    "report_date", "brand", "category_l1", "total_views",
    "total_purchases", "revenue", "conversion_rate_pct"

]])
```

10. LIMITATIONS & RISKS

- Single-node DuckDB (it is not a multi-user cloud warehouse).
- For large volumes, DuckDB may require disk spill and sufficient free disk space.
- conversion_rate_pct can be NULL when total_views = 0.
- fact_events is deduplicated by event_id; if source event semantics change, a full refresh may be required.
- Kafka and CSV share the same bronze layer, so correctness depends on consumer offsets and deduplication rules.

11. GOVERNANCE & CONTACT

Owner: Alicja Tomaszewska, s35133

Contact: s35133@pjwstk.edu.pl

Steward / responsible person: Alicja Tomaszewska

Access policy: Public for class

Marketplace location: MS Teams / Data Marketplace folder

Feedback